



THE EXPLOSIVE SUBSTANCES — FULL MIND MAP

Chemistry · Chapter 7.E · Topic-wise · Fully Explained · Hinglish · SSC/PCS/UPSC Ready

Topics in this chapter:

1. Mine Explosions & Non-Explosive Trap
2. Nitroglycerine, Nobel Oil & Dynamite
3. RDX: Discovery, Names & Key Facts
4. HMX / Octogen — High Explosive
5. Cordite, Blasting Gelatine & Amatol
6. Quick Revision — High-Yield Explosive Pairs

1. Mine Explosions & Non-Explosive Trap

- Coal/mines me methane (CH_4) air ke saath explosive mixture bana sakta hai. Source ke mutabik approx. 5–15% methane-in-air range especially dangerous/explosive hoti hai; given options me mine explosions ka standard answer Methane + Air hai. [Q1]
- Nitrochloroform = Chloropicrin ko source “not an explosive” ke roop me treat karta hai; iska use historically toxic/tear-gas aur agriculture me fumigant/fungicidal-insecticidal chemical ke roop me hua hai. TNT, TNG/RDX aur Potassium chlorate explosive contexts me aate hain. [Q2–Q3]

2. Nitroglycerine, Nobel Oil & Dynamite

- Nitroglycerine ko Trinitroglycerol (TNG), Glyceryl trinitrate (GTN) aur Nobel oil bhi kaha jata hai. Ye heavy, oily, highly energetic liquid explosive hai. [Q4, Q9]
- Glycerol ko concentrated $\text{HNO}_3 + \text{H}_2\text{SO}_4$ ke saath nitrate karne par Nitroglycerine/Glycerol trinitrate banta hai; isi wajah se Glycerol explosive manufacture ka important precursor hai. [Q11]
- Dynamite ka active explosive chemical Nitroglycerine (Glycerol trinitrate) hota hai; practical dynamite me isse absorbent/stabilising material ke saath safer handling ke liye formulate kiya jata hai. [Q12]

⚠ EXAM TRAP — Q2–Q3 — Nitrochloroform / Chloropicrin

- ▶ Source ise “not an explosive” answer karta hai; standard exam context me Chloropicrin ko primarily fumigant/chemical-warfare irritant ke roop me yaad rakho.
- ▶ Potassium chlorate khud high explosive ke roop me normal handling material nahi hai, lekin strong oxidiser hone ke karan explosive mixtures/pyrotechnic compositions me use ho sakta hai. Question ka option-context important hai.

3. RDX: Discovery, Names & Key Facts

- RDX ek powerful nitroamine high explosive hai. George Friedrich Henning ne is compound ko 1890s me prepare/patent kiya; exam source inventor/discoverer ke roop me Henning ko mark karta hai. [Q5]
- RDX ka chemical name Cyclotrimethylenetrinitramine hai; common alternate names Cyclonite, Hexogen aur T4 hain. Source formula ko $(\text{CH}_2\text{N}_2\text{O}_2)_3$ ke form me deta hai. [Q6–Q7]
- RDX TNT se zyada powerful high explosive maana jata hai aur World War II ke dauran military use me widely aaya. Exam pair yaad rakho: RDX → Cyclonite/Hexogen. [Q6–Q7]
- RDX ko “Research Department Explosive” naam se commonly expand kiya jata hai; kuch references “Royal Demolition Explosive” bhi likhte hain, isliye objective exam me given source wording ko follow karo. [Q6]

4. HMX / Octogen — High Explosive

- HMX ko Octogen bhi kaha jata hai; ye RDX se chemically related powerful aur comparatively insensitive nitroamine high explosive hai. Exam source HMX ka full form “High Melting Explosive” accept karta hai. [Q8]
- HMX naam ki expansion literature me multiple forms me mil sakti hai, lekin given question me only statement 1 correct hai. Isliye option “Only 1” source-key ke according sahi hai. [Q8]

⚠ EXAM TRAP — Q8 — HMX ko “colourless liquid” mat yaad karo

- ▶ Source explanation me HMX ko colourless liquid likha gaya hai, lekin HMX/Octogen actually crystalline solid high explosive hota hai.
- ▶ Exam key ka relevant part full form/identity hai: HMX → High Melting Explosive (accepted source answer) aur alternate name Octogen.

5. Cordite, Blasting Gelatine & Amatol

- Amatol me Nitroglycerine essential constituent nahi hota. Ye mainly Ammonium nitrate + TNT ka mixture hota hai; isi se Cordite/Blasting Gelatine/Dynamite se differentiate kiya jata hai. [Q10]
- Cordite aur Blasting Gelatine traditional formulations me Nitroglycerine-containing explosives/propellants rahe hain; Dynamite bhi Nitroglycerine-based hota hai. [Q10]
- High-yield distinction: Nitroglycerine-containing group → Cordite, Blasting Gelatine, Dynamite; Nitroglycerine-free given option → Amatol. [Q10]



EXAM TRAP — Q10 — “Amatol is an explosive element” wording wrong hai

- ▶ Amatol koi element nahi; ye Ammonium nitrate aur TNT ka explosive mixture hai.
- ▶ Question ka tested fact bas itna hai ki Amatol me Nitroglycerine essential constituent nahi hota.

6. Quick Revision — High-Yield Explosive Pairs

- Fast revision pairs: Mine explosion → Methane + Air; Nobel oil/TNG → Nitroglycerine; RDX → Cyclonite/Hexogen; HMX → Octogen; Dynamite active chemical → Glycerol trinitrate; Amatol → Ammonium nitrate + TNT. [Q1–Q12]

✓ Source-question mapping preserved: Q1–Q12